

Ethan Wu

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EDUCATION

Queen's University

Bachelor of Applied Science, Mechatronics and Robotics Engineering

– GPA: 3.89/4.3 | Dean's Scholar

Kingston, ON

Sept. 2024 – April 2028

EXPERIENCE

Software Lead – NASA HERC Rover

Sept. 2026 – Present

Queen's Lunar Exploration Team

Kingston, ON

- Architecting the software stack for a 6-wheeled rocker-bogie autonomous rover in ROS2 and C++ for the 2027 NASA HERC competition.
- Designing a radio command pipeline to decode real-time VHF telemetry and drive state machine command sequences.
- Developing OpenCV target recognition algorithms to isolate geometric shapes, compute target distances via focal geometry, and dynamically apply digital color filters.
- Configuring rocker-bogie kinematic differential steering models and Nav2 SLAM pipelines for autonomous terrain navigation.

Embedded Systems Member

Sept. 2025 – June 2026

Queen's AutoDrive Team

Kingston, ON

- Engineered a C++ memory monitoring module for an embedded watchdog to detect real-time system crashes.
- Evaluated system failure probabilities and active redundancy models to improve compute node fault tolerance.
- Earned 2nd Place in the Mobility Innovation Challenge at the SAE AutoDrive Competition for an AV adoption framework.

Makerspace Volunteer Supervisor

Sept. 2026 – Present

SparQ Makerspace, Queen's University

Kingston, ON

- Supervise prototyping, providing technical instruction on FDM 3D printing, laser cutting, and circuit soldering.
- Maintain benchtop diagnostic equipment, power tools, and lab machinery while enforcing shop safety standards.

Automation Team Manager

Oct. 2025 – Present

Queen's Engineering Society

Kingston, ON

- Lead a development team constructing automated Power Automate flows and web interfaces to streamline organization data pipelines.
- Manage technical developer recruitment, project onboarding, and system architecture for administrative tools.

PROJECTS

TriPAD - Construction Site Sensor | C++, ESP32, Sensor Fusion

Apr. 2026 – Aug. 2026

- Built a standalone environmental monitoring unit using an ESP32 to process particulate (PMS5003), noise (MAX9814), and IMU vibration (MPU6050) sensor telemetry over I2C and UART.
- Designed a 3D-printed Autodesk Fusion enclosure and won 3rd place at the General's Lab Innovation competition.

Mars Sample Return (MSR) Rover | ROS2, C++, Python, R&D

Jan. 2026 – Apr. 2026

- Developed a ROS2-based autonomous rover with a bidirectional Raspberry Pi-to-Arduino communication bridge.
- Implemented closed-loop PI control with IMU/encoder sensor fusion and integrated RPLIDAR A1 for Nav2 localization.

Autonomous Transforming Robot | Autodesk Fusion, Kinematics

Jan. 2026 – Present

- Designing a 30+ DOF kinematic assembly with interference checks, motion analysis, and tolerance tuning for 3D printing.

TECHNICAL SKILLS

Software & Controls: C/C++, Python, ROS2, Nav2, SLAM, OpenCV, PID/PI Control, Sensor Fusion, Kinematics, SQL, Linux

Hardware & Embedded: ESP32, Raspberry Pi, Arduino, CAN bus, UART, I2C, PMS5003, MAX9814, MPU6050, LiDAR, Encoders

CAD & Tools: Autodesk Fusion, SOLIDWORKS, LTSpice, Quartus, MATLAB, MS Power Automate, Git, LaTeX

Hands-On: Soldering, Circuit Debugging, Machining, Welding, Woodworking, 3D Printing; Languages: Cantonese, Mandarin, French